# INTRODUCTION TO PROBABILITY

STAT 2A: APPLIED STATISTICS
MIDYEAR, A.Y. 2024-2025

# TERMS AND DEFINITION

## EXPERIMENT

It refers to any process that generates a set of data. It can be classified into three types:

- Designed experiments
- Observational studies
- Retrospective studies

# TERMS AND DEFINITION

## SAMPLE SPACE

It is a set containing all the possible outcomes of a given statistical experiment. It is denoted by the capital letter “S”.

*Example:*

**Experiment:** Tossing a fair coin once

**Sample Space:** $S = \{Head, Tail\}$

**Experiment:** Rolling a regular die once

**Sample Space:** $S = \{1,2,3,4,5,6\}$

# TERMS AND DEFINITION

## SAMPLE POINT

It refers to each element or component making up the sample space.

# TERMS AND DEFINITION

## EVENT

It refers to any subset formed from a sample space. It is typically denoted using any capital letter.

*Example:*

***Suppose a coin is tossed three times, define the following events:***

1. Event A: only one tail will show up
2. Event B: at most two heads will show up
3. Event C: at least one tail will show up

## ***Suppose a coin is tossed three times, define the following events:***

The *sample space* for this experiment can be defined as:

$$\boldsymbol{S = \{HHH, HHT, HTH, THH, TTH, THT, HTT, TTT\}}$$

1. Event A: only one tail will show up

$$A = \{HHT, HTH, THH\}$$

2. Event B: at most two heads will show up

$$B = \{HHT, HTH, THH, TTH, THT, HTT, TTT\}$$

3. Event C: at least one tail will show up

$$C = \{HHT, HTH, THH, TTH, THT, HTT, TTT\}$$

# TERMS AND DEFINITION

## COMPLEMENT OF AN EVENT

Suppose $A$ is an event under $S$, the complement of $A$ is the set of all sample points that are not in an event (A). In symbols, it can be expressed as $\boldsymbol{A'}$ or $\boldsymbol{\sim A}$.

## UNION OF EVENTS

Suppose $A$ and $B$ are two events under $S$, union of events $A$ and $B$ is the set of all sample points that are either in $A$ only, in $B$ only, or in both $A$ and $B$. In symbols, it can be expressed as $\boldsymbol{A \cup B}$.

# TERMS AND DEFINITION

## INTERSECTION OF EVENTS

Suppose $A$ and $B$ are two events under $S$, intersection of events $A$ and $B$ is the set of all sample points that are common to both $A$ and $B$. In symbols, it can be expressed as $\boldsymbol{A \cap B}$**.**

## MUTUALLY EXCLUSIVE/DISJOINT EVENTS

Suppose $A$ and $B$ are two events under $S$, $A$ and $B$ is are said to be mutually exclusive if and only if their intersection is a *null (empty) set*.

# COUNTING TECHNIQUES

# DEFINITION OF COUNTING TECHNIQUES

- In Probability, counting techniques refer to various methods of determining the number of sample points formed in a given statistical experiment.
- It is important to note that these counting techniques are used to identify *how many* sample points are there and not what specifically the sample points are.

# TYPES OF COUNTING TECHNIQUES

- Fundamental Principle of Counting
- Permutation
- Combination

# FUNDAMENTAL PRINCIPLE OF COUNTING

If a certain experiment is performed with a series of steps or operations such that the first step can be performed in $n_1$ ways, the second step in $n_2$ ways, and so on, then, the number of possible ways or outcomes in the experiment can be computed as

$$n_1 \times n_2 \times n_3 \ldots \times n_k$$

where $k$ is the number of steps or operations involved.

# FUNDAMENTAL PRINCIPLE OF COUNTING

ILLUSTRATIVE EXAMPLE

**1. How many sample points are there if a pair of fair dice is rolled?**

Manually, we can find the answer through systematic listing. But for larger and more complex experiments, listing is not a practical method.

| | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| **1** | 1 , 1 | 1 , 2 | 1 , 3 | 1 , 4 | 1 , 5 | 1 , 6 |
| **2** | 2 , 1 | 2 , 2 | 2 , 3 | 2 , 4 | 2 , 5 | 2 , 6 |
| **3** | 3 , 1 | 3 , 2 | 3 , 3 | 3 , 4 | 3 , 5 | 3 , 6 |
| **4** | 4 , 1 | 4 , 2 | 4 , 3 | 4 , 4 | 4 , 5 | 4 , 6 |
| **5** | 5 , 1 | 5 , 2 | 5 , 3 | 5 , 4 | 5 , 5 | 5 , 6 |
| **6** | 6 , 1 | 6 , 2 | 6 , 3 | 6 , 4 | 6 , 5 | 6 , 6 |

# FUNDAMENTAL PRINCIPLE OF COUNTING

Based on the given experiment, there are two (2) components to the whole experiment:

1. The number of possibilities on the first die $(n_1)$; and
2. The number of possibilities on the second die $(n_2)$

To determine the number of possible outcomes for the whole experiment, we just need to multiply the number of possible outcomes at each component/step:

Total number of outcomes $= n_1 \times n_2$

$= 6 \times 6$

$\boxed{= 36 \text{ outcomes}}$

# FUNDAMENTAL PRINCIPLE OF COUNTING

<u>ILLUSTRATIVE EXAMPLE</u>

**2. Sam plans to assemble a computer by himself. He has the choice of ordering CPU from two brands, hard drive from four, memory card from three, and accessory bundle from five stores. In how many ways can he order the parts for his computer if he only needs one of each type?**

$$\begin{aligned}\text{Total number of ways} &= n_1 \times n_2 \times n_3 \times n_4 \\ &= 2 \times 4 \times 3 \times 5 \\ &\boxed{= \mathbf{120\ ways}}\end{aligned}$$

# PERMUTATION

This counting technique is used to determine the **number of possible arrangements** that can be formed if all or some parts of the set is considered. There are various ways permutation can be computed depending on the context of the experiment given.

# PERMUTATION

## 1. Linear Permutation of all $n$ distinct objects

The linear permutation of $n$ objects all taken at the same time is $n!$ (read as "n-factorial")

*Example 1:*

**In how many ways can the letters A, B, C, & D be arranged to form a 4-letter code with no repetitions?**

$$4! = 4 \times 3 \times 2 \times 1$$
$$\boxed{= \mathbf{24\ ways}}$$

# PERMUTATION

| | | | |
|---|---|---|---|
| ABCD | BACD | CABD | DABC |
| ABDC | BADC | CADB | DACB |
| ACBD | BCAD | CBAD | DBAC |
| ACDB | BCDA | CBDA | DBCA |
| ADBC | BDAC | CDAB | DCAB |
| ADCB | BDCA | CDBA | DCBA |

# PERMUTATION

*Example 2:*

Joshua wants to create a password for his newly created Facebook account. He wants to use all the letters of his name (in lowercase), as well as the digits 0, 9, 1, and 7 which corresponds to his birthdate (September 17). How many distinct passwords can he choose from if he wants to use all the mentioned letters and numbers?

$$10! = 10 \times 9 \times 8 \times \cdots \times 3 \times 2 \times 1$$

$$\boxed{= \mathbf{3\,628\,800\ ways}}$$

# PERMUTATION

## 2. Linear Permutation of $n$ distinct objects taken $r$ at a time

The linear permutation of $n$ objects taken $r$ at a time is:

$$_{n}\mathrm{P}_{r} = \frac{n!}{(n-r)!}$$

*Example 3:*

**How many distinct pairs can be formed using the letters a, b, c, d, & e such that no letter is paired with itself?**

# PERMUTATION

*Example 3:*

**How many distinct pairs can be formed using the letters a, b, c, d, & e such that no letter is paired with itself?**

$$_{n}P_{r} = \frac{n!}{(n-r)!}$$

$$_{5}P_{2} = \frac{5!}{(5-2)!}$$

$$= \frac{5\times4\times3\times2\times1}{3\times2\times1}$$

$$= \frac{120}{6}$$

$$\boxed{= \mathbf{20\ ways}}$$

# PERMUTATION

*Example 3:*

**How many distinct pairs can be formed using the letters a, b, c, d, & e such that no letter is paired with itself?**

| | | | | |
|---|---|---|---|---|
| AB | BA | CA | DA | EA |
| AC | BC | CB | DB | EB |
| AD | BD | CD | DC | EC |
| AE | BE | CE | DE | ED |

# PERMUTATION

*Example 4:*

**During the grand finals of Bb. CvSU 2025, seven grand finalists are left and only three awards are left (Bb. CvSU 2025, 1st Runner-up, and 2nd Runner-up). In how many ways can the final ranking turn out?**

$$_{n}P_{r} = \frac{n!}{(n-r)!}$$

$$_{7}P_{3} = \frac{7!}{(7-3)!}$$

$$= \frac{7\times6\times5\times4\times3\times2\times1}{4\times3\times2\times1}$$

$$= \frac{5\ 040}{24}$$

$$\boxed{= \mathbf{210\ ways}}$$

# PERMUTATION

## 3. Permutation of $n$ distinct objects arranged in a circle

The number of distinct arrangements of $n$ distinct objects arranged in a circle is computed as $(n-1)!$

*Example 5:*

**Six students are about to play "spin-the-bottle". In how many ways can they arrange themselves for the game?**

$$P_n = (n-1)!$$
$$P_6 = (6-1)!$$
$$= 5!$$
$$\boxed{= 120 \text{ ways}}$$

# PERMUTATION

*Example 6:*

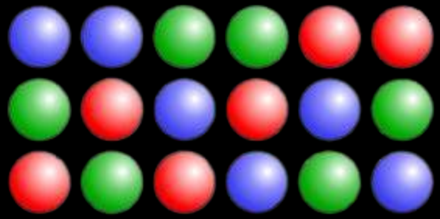
**Suppose that from the six students who are about to play "spin-the-bottle", two have requested to be seated next to each other. In how many ways can they now arrange themselves?**

This is the number of elements that are "not free" and forms a separate group

$$
\begin{aligned}
P &= (6 - 2)! \times 2! \\
&= 4! \times 2! \\
&= 24 \times 2 \\
&\boxed{= \mathbf{48\ ways}}
\end{aligned}
$$

This is still based on the number of elements that are "not free".

Its permutation is evaluated to denote that the elements inside this group can also be rearranged within themselves



# PERMUTATION

## 4. Permutation with non-distinct objects

There are instances that arrangements needed to be formed contain objects that are non-distinct or has replicates. In these instances, we compute the permutation by considering the how the objects are duplicated such that $n_1$ is of one kind, $n_2$ is of the second kind, $n_3$ is of the third kind, and so on until $n_k$ which is of the $k$th kind. Furthermore, the number of arrangements is computed as:

$$\mathrm{P} = \frac{n!}{n_1! \times n_2! \times \ldots \times n_k!}$$

# PERMUTATION

*Example 7:*

**Karina plans to arrange her books in her newly-bought bookshelf. She has a total of 10 books which are as follows: 2 mystery books, 4 romance books, 3 sci-fi books, and 1 fantasy book. In how many ways can she arrange the books if only the genres are to be distinguished?**

$$\text{No. of Permutations} = \frac{n!}{n_1! \times n_2! \times \ldots \times n_k!}$$

$$= \frac{10!}{2! \times 4! \times 3! \times 1!}$$

$$= \frac{3\ 628\ 800}{2 \times 24 \times 6 \times 1}$$

$$= \frac{3\ 628\ 800}{288}$$

$$\boxed{= \mathbf{12\ 600\ ways}}$$

# PERMUTATION

*Example 8:*

**For a science project, Luke needs to create a LED strip made up of various LED lamps. If he has 5 red lamps, 3 blue lamps, and 4 green lamps, in how many ways can he design his project?**

$$\text{No. of Permutations} = \frac{n!}{n_1! \times n_2! \times \ldots \times n_k!}$$

$$= \frac{12!}{5! \times 3! \times 4!}$$

$$= \frac{479\,001\,600}{120 \times 6 \times 24}$$

$$= \frac{479\,001\,600}{17\,280}$$

$$\boxed{= \mathbf{27\,720\ ways}}$$

# PERMUTATION

## 5. Permutation involving partitioning of $n$ objects into $k$ cells

There are instances that a set of $n$ objects are to be partitioned into $k$ groups (or cells) with different dimensions $(k_1, k_2, \ldots, k_i)$. In these cases, to compute for the number of ways the partitioning can be performed:

$$\mathrm{P} = \frac{n!}{k_1! \times k_2! \times \ldots \times k_i!}$$

# PERMUTATION

*Example 9:*

**In how many ways can seven tourists be assigned to Room A** (a triple-bed room), **Room B** (a double-bed room), **and Room C** (a double-bed room)**?**

$$\text{No. of Permutations} = \frac{n!}{k_1! \times k_2! \times \ldots \times k_i!}$$

$$= \frac{7!}{3! \times 2! \times 2!}$$

$$= \frac{5\,040}{6 \times 2 \times 2}$$

$$= \frac{5\,040}{24}$$

$$\boxed{= 210 \text{ ways}}$$

# PERMUTATION

*Example 10:*

**In how many ways can twelve people be grouped into groups of three members each assigned with unique roles** (leader, asst. leader, secretary)**?**

$$\text{No. of Permutations} = \frac{n!}{k_1! \times k_2! \times \ldots \times k_i!}$$

$$= \frac{12!}{3! \times 3! \times 3! \times 3!}$$

$$= \frac{479\,001\,600!}{6 \times 6 \times 6 \times 6}$$

$$= \frac{479\,001\,600}{1\,296}$$

$$\boxed{= 369\,600 \text{ ways}}$$

# COMBINATION

This counting technique is used to determine the number of possible selections that can be formed if all or some parts of the set is considered **regardless of the arrangement or order**. The general formula in computing for the combination of $n$ objects taken $r$ at a time is:

$${}_{n}\mathrm{C}_{r} = \frac{n!}{r!\,(n-r)!}$$

# COMBINATION

*Example 1:*

**A boy was asked by his mother to randomly get three shirts from a collection of nine shirts in his mother's closet. In how many ways can the young boy perform the selection?**

$$_nC_r = \frac{n!}{r!\,(n-r)!}$$

$$_9C_3 = \frac{9!}{3!\,(9-3)!}$$

$$= \frac{9\times8\times7\times6\times5\times4\times3\times2\times1}{(3\times2\times1)(6\times5\times4\times3\times2\times1)}$$

$$= \frac{362\,880}{4\,320}$$

$$\boxed{= \mathbf{84\ ways}}$$

# COMBINATION

*Example 2:*

**Jenny has a padlock as shown in the photo on the right. Due to some reason, she forgot what her 4-digit lock combination was and needs to try each combination one-by-one. What is the maximum number of 4-digit combinations she needs to try in order to open the lock?**

# COMBINATION


$$
\begin{aligned}
{}_{n}C_{r} &= \frac{n!}{r!\,(n-r)!} \\
{}_{10}C_{4} &= \frac{10!}{4!\,(10-4)!} \\
&= \frac{10\times9\times8\times7\times6\times5\times4\times3\times2\times1}{(4\times3\times2\times1)(6\times5\times4\times3\times2\times1)} \\
&= \frac{3\,628\,800}{(24)(720)} \\
&= \frac{3\,628\,800}{17\,280} \\
&\boxed{= \textbf{210 combinations}}
\end{aligned}
$$

# WRITTEN WORK #4

Read and answer each problem carefully. Write your complete computation and box your final answer.

1. In how many ways can a student answer a 5-item "True or False" quiz if none of the items will be left unanswered?
2. In a movie theater, five couples are to be seated in a row. In how many ways can they be arranged if every couple must be seated beside their respective partners?
3. A family of eight people is to be seated on a long dining table. In how many ways can they take their seats if the parents must be seated on both ends of the table?
4. How many "full house" poker hand can be made out of a standard 52-card deck? Note: a full house is a 5-card combination made up of a trio (three of a kind) and a pair (two cards of the same rank)
5. How many possible standard plate numbers can be formed using five letters (A, B, C, D, E) and three numbers (5, 7, 9) if each plate is made up of three letters followed by three numbers and repetition of letters and numbers is allowed?